आरंभ

PAIR OF LINEAR EQUATIONS IN TWO VARIABLES

Lecture #5

#AIK :

$\boxed{\text{PLE2V}}$

- Graphical
- Algebraic Method
  - Substitution
  - Elimination.

Word Problems:

#LP: The cost of 4 pens and 4 pencil boxes is Rs. 100. Three times the cost of a pen is Rs. 15 more than the cost of a pencil box. Form the pair of linear equations for the above situation. Find the cost of a pen and a pencil box.

$\boxed{(4 \text{ pen}) + (4 \text{ pencil box}) = 100}$

Let cost of a pen = ₹ x

" " " P.B = ₹ y

$4x + 4y = 100$

$\cancel{4}(x+y) = \cancel{100}\,25$

$\Rightarrow \boxed{x + y = 25}$ — (II)

$\boxed{3 \times (\text{cost of a pen}) = 15 + (\text{cost of 1 PB})}$

$3 \times (x) = 15 + y$

$3x = 15 + y$

$\boxed{3x - y = 15}$ — (I)

PLE2V

$$\begin{cases} x + \cancel{y} = 25 \\ 3x - \cancel{y} = 15 \end{cases}$$

$+$

$\cancel{4}x = \cancel{40}$ 10

put

$\boxed{x = 10}$

$\Downarrow$

$10 + y = 25$

$\boxed{y = 15}$

Final Ans

Cost of a pen ⇒ ₹x ⇒ ₹10

Cost of a Pencilbox ⇒ ₹y ⇒ ₹15.

#LP: 7 audio cassettes and 3 video cassettes cost ₹ 1110, while 5 audio cassettes and 4 video cassettes cost ₹ 1350. Find the cost of audio cassettes and video cassettes.

Let cost of 1 A.C = ₹ x

cost of 1 V.C = ₹ y

ATQ $\boxed{7x + 3y = 1110}$ — (I)

ATQ $\boxed{5x + 4y = 1350}$ — (II)

$7x + 3y = 1110$

$$5x + 4y = 1350 \quad \times 3 \longrightarrow 15x + \cancel{12y} = 4050$$

$$7x + 3y = 1110 \quad \times 4 \longrightarrow \underset{-}{28x} \underset{-}{+} \cancel{12y} = \underset{-}{4440}$$

$$\cancel{+13}x = +\cancel{390}^{30}$$

$$\boxed{x = 30}$$

put

$\Downarrow$

$$7(30) + 3y = 1110$$

$$210 + 3y = 1110$$

$$3y = 1110 - 210$$

$$\cancel{3}y = \cancel{900}^{300}$$

$$y = 300$$

$$\left.\begin{array}{l} 1\ A \cdot C \Rightarrow ₹x \Rightarrow \underline{₹30} \\ 1\ V \cdot C \Rightarrow ₹y \Rightarrow \underline{₹300} \end{array}\right\} \underline{\text{Ans.}}$$

#LP: Reena has pens and pencils, which together are 40 in number. If she has 5 more pencils and 5 less pens, then the number of pencils would become 4 times the number of pens. Find the original number of pens and pencils.

Before

Let no. of pen $\Rightarrow x$ pens

no. of pencils $\Rightarrow y$ pencils

ATQ $\boxed{x + y = 40}$ — (I)

After (ATQ)

$(\text{pen})' \Rightarrow (x-5)$

$(\text{pencil})' \Rightarrow (y+5)$

ATQ $(\text{pencil})' = 4 \times (\text{pen})'$

$y + 5 = 4 \times (x-5)$

$y + 5 = 4x - 20$

$5 + 20 = 4x - y$

$\boxed{4x - y = 25}$ — (II)

$x = 13$ pens

$y = 27$ pencils

sum = 180.

#LP: The larger of two supplementary angles exceeds the smaller by 18°. Find the angles. (CBSE 2019)

Let larger angle is x°
smaller angle is y° $(x^\circ > y^\circ)$

ATQ

Larger = smaller + 18

$x = y + 18$

$\boxed{x - y = 18}$ — (I)

$\boxed{x + y = 180}$ — (II)

$x = 99^\circ$
$y = 81^\circ$ Ans

#LP: In a ΔABC, $\angle A = x^0$, $\angle B = (3x-2)^0$, $\angle C = y^0$. Also, $\angle C - \angle B = 9^0$. Find the three angles.

ATQ

$\angle C - \angle B = 9$

$y - (3x-2) = 9$

$y - 3x + 2 = 9$

$\boxed{y - 3x = 7}$ — (I)

for a $\Delta \rightarrow$ sum of all angles $= 180$

$\angle A + \angle B + \angle C = 180$

$x + (3x-2) + y = 180$

$\boxed{4x + y = 182}$ — (II)

$x = 25$

$y = 82$

Final Ans:

$\angle A = x = 25^\circ$

$\angle B = 3x-2 \Rightarrow 3(25)-2 \Rightarrow 73^\circ$

$\angle C = y = 82^\circ$

K3B

Tshirt factory

Cost Price of 1 T-shirt ⇒ ₹100

$\boxed{C.P = ₹100}$

Selling price (S.P) ⇒ 100 + (25% of 100)

C.P. + Profit

$\Rightarrow 100 + \frac{25}{\cancel{100}} \times \cancel{100}$

$\boxed{S.P. \rightarrow ₹125}$

LP:- If the cost price of a toy car is ₹550. And the owner wants to make 35% profit. Then what will be the selling price of that toy car.

$C.P = ₹550$

$S.P = C.P + \text{Profit}$

$= 550 + (35\% \text{ of } 550)$

$= 550 + \frac{35}{\cancel{100}_{2}} \times \cancel{550}^{11}$

$= 550 + \frac{\cancel{385}}{\cancel{2}} 192.5$

$= 742.5$ Rs

#LP: Jamila sold a table and a chair for ₹ 1050, thereby making a profit of 10% on the table and 25% on the chair. ... a ... t ... Find the cost price of each.

Let C.P of a table $\Rightarrow$ ₹ x

" " " chair $\Rightarrow$ ₹ y

CASE . I

table $\Rightarrow$ 10%

chair $\Rightarrow$ 25%

$S.P \Rightarrow CP + \text{profit}$

$\Rightarrow x + (10\% \text{ of } x)$

$\Rightarrow x + \frac{10x}{100}$

$\Rightarrow x + \frac{x}{10}$

$\Rightarrow \frac{10x + x}{10} \Rightarrow \frac{11x}{10} = \text{SP of table}$

$SP = C.P + \text{Profit}$

$= y + \frac{25}{100}y$

$= y + \frac{y}{4}$

$\Rightarrow \frac{4y + y}{4} \Rightarrow \frac{5y}{4}$ SP of chair

$\boxed{\text{SP table} + \text{SP chair} = 1050}$

$\Rightarrow \frac{11x}{10} + \frac{5y}{4} = 1050$

$\Rightarrow \frac{22x + 25y}{20} = 1050$

$\Rightarrow 22x + 25y = 1050 \times 20$

$\boxed{22x + 25y = 21000}$ (I)

#LP: Jamila sold a table and a chair for [redacted] If she had taken a profit of 25% on the table and 10% on the chair, she would have got ₹ 1065. Find the cost price of each.

Let C.P of a table ⇒ ₹ x

" " " chair ⇒ ₹ y

Case (II):-

Table ⇒ 25%

$SP = CP + \text{profit}$

$= x + \frac{\cancel{25}}{\cancel{100}}x$

$= x + \frac{x}{4}$

$\boxed{SP_{table} = \frac{5x}{4}}$

chair = 10%

$SP = \text{cost} + \text{profit}$

$= y + \frac{\cancel{10}}{\cancel{100}}y$

$= y + \frac{y}{10}$

$SP_{chair} = \frac{11y}{10}$

$SP_{table} + SP_{chair} = 1065$

$\Rightarrow \frac{5x}{4} + \frac{11y}{10} = 1065$

$\Rightarrow \frac{25x + 22y}{20} = 1065$

$\Rightarrow 25x + 22y = 1065 \times 20$

$\boxed{\Rightarrow 25x + 22y = 21300}$ (II)

$$\text{I} \longrightarrow 22x + 25y = 21000$$

$$\text{II} \longrightarrow 25x + 22y = 21300$$

Add

$$22x + 25y = 21000$$
$$\oplus\ 25x + 22y = 21300$$
$$47x + 47y = 42300$$
$$\cancel{47}(x+y) = \cancel{42300}\ 900$$
$$\boxed{x + y = 900} - \text{A}$$

Subtract

$$22x + 25y = 21000$$
$$-\ 25x + 22y = 21300$$
$$-3x + 3y = -300$$
$$\cancel{-3}(x-y) = \cancel{-300}\ 100$$
$$\boxed{x - y = 100} - \text{B}$$

$$\boxed{\begin{array}{l} x = ₹5000 \\ y = ₹4000 \end{array}}$$

$x = ₹5000$ ← cp of table

$y = ₹4000$ ← cp of chair

#LP: Two schools 'P' and 'Q' decided to award prizes to their students for two games of Hockey Rs. x per student and Cricket Rs. y per student. School 'P' decided to award a total of Rs. 9,500 for the two games to 5 and 4 students respectively, while school 'Q' decided to award Rs. 7,370 for the two games to 4 and 3 students respectively. Based on the given information, answer the following questions. (CBSE 2023)

(i) Represent the following information algebraically (~~in terms of x and y~~).

Hockey ⇒ 1 student = ₹ x

Cricket ⇒ 1 student = ₹ y

P

$$\boxed{5x + 4y = 9500} \quad \text{— (I)}$$

Q

$$\boxed{4x + 3y = 7370} \quad \text{— (II)}$$

(ii) (a) What is the prize amount for hockey? OR (b) Prize amount on which game is more and by how much?

$y > x$

cricket has more prize than hockey by ₹170

$x \Rightarrow$ ₹980

$y \Rightarrow$ ₹11.50

(iii) What will be the total prize amount if there are 2 students each from two games?

Total $\Rightarrow$ 2 hockey + 2 cricket

$\Rightarrow$ ₹$(2 \times 980 + 2 \times 1150)$ A

Next Toppers

HW → Module last class Q.

+ class revise

THANK YOU COODIESS !!